Lecture 20: Solving LCCDE using Z transform

- Recall from 2714 that a linear, constant coefficient difference equation is a relationship between shifted versions of input and output.

  - Advance form $\sum_{k=0}^{N} a_k y[n+k] = \sum_{k=0}^{M} b_k x[n+k]$

    e.g. $2y[n+1] - y[n] = x[n+1]$ $\quad N = M = 1$

  - Delay form $\sum_{k=0}^{N} a_k y[n-P+k] = \sum_{k=0}^{M} b_k x[n-P+k]$ $\quad P = \max(N, M)$

  - Recursive form $y[n] = -\sum_{k=1}^{N} \frac{a_k}{a_0} y[n-P+k] + \sum_{k=0}^{M} \frac{b_k}{a_0} x[n-P+k]$

- Given $x[n]$ and $N+M$ auxiliary/initial conditions we wish to solve the Difference Equation, i.e. find expression for $y[n]$

The solution can be divided into two parts

  - Zero-input due to I.C. only $y_{zi}[n]$
    also called natural response

  - Zero-state due to input only $y_{zs}[n]$ also called forced response.

Thus $y[n] = y_{zi}[n] + y_{zs}[n]$

When the I.C. are zero $y[n] = y_{zs}[n]$ and the system is LTI.

- The time/index shift property gives us a straightforward approach to solving LCCDE's.

- Recall the index shift property of z transform is

$$x[n] \overset{z}{\longleftrightarrow} X(z)$$

- Delay (right shift)

$$x[n-k]u[n-k] \overset{z}{\longleftrightarrow} z^{-k}X(z)$$

$$x[n-k]u[n] \longleftrightarrow z^{-k}X(z) + \underbrace{z^{-k}\sum_{n=1}^{k} x[-n]z^{n}}_{\text{z transform of k values before } n=0}$$

$k>0$

- Advance (left shift)

$$x[n+k]u[n] \longleftrightarrow z^{k}X(z) - \underbrace{z^{k}\sum_{n=0}^{k-1} x[n]z^{-n}}_{\text{z transform of k-1 values from } n=0\ k-1}$$

$k>0$

For Delay form we need k values before $n=0$ as A.C.
" Advance form " " " " after $n=-1$ as A.C.

which form we use depends on A.C. given (before $n=0$, after $n=0$) typically delay or recursive form.

- Example: Given $y[n+1] + ay[n] = x[n+1]$ and $y[-1] = B$

Then rewrite in delay form

$$y[n] + ay[n-1] = x[n]$$

Take z transform $Y(z) + a\left[z^{-1}Y(z) + y[-1]\right] = X(z)$

Solve for $Y(z)$ $\Rightarrow$ $Y(z)(1+az^{-1}) = -ay[-1] + X(z)$

$$Y(z) = \frac{-ay[-1]}{1+az^{-1}} + \frac{X(z)}{1+az^{-1}}$$

$$= \underbrace{\frac{-ay[-1]z}{z+a}}_{Y_{zi}(z)} + \underbrace{\frac{z}{z+a}X(z)}_{Y_{zs}(z)}$$

$$Y_{zs}(z) = H(z)Y(z)$$

$H(z)$: transfer function

– Example cont. Suppose $x[n] = (\frac{1}{2})^n u[n]$ and $B = 1$

then $Y(z) = \dfrac{-az}{z+a} + \dfrac{z}{z+a} \cdot \dfrac{z}{z-\frac{1}{2}}$

$$y_{zi}[n] = Z^{-1}\left\{\frac{-az}{z+a}\right\} = -a(-a)^n u[n]$$

$$y_{zs}[n] = Z^{-1}\left\{\frac{z^2}{(z+a)(z-\frac{1}{2})}\right\}$$

Two cases: $a \neq -\frac{1}{2}$ Then $\dfrac{Y_{zs}(z)}{z} = \dfrac{A}{z+a} + \dfrac{B}{z-\frac{1}{2}}$

$$A = \left.\frac{z}{z-\frac{1}{2}}\right|_{z=-a} = \frac{-a}{-a-\frac{1}{2}} = \frac{2a}{2a+1}$$

$$B = \left.\frac{z}{z+a}\right|_{z=\frac{1}{2}} = \frac{1/2}{1/2+a} = \frac{1}{2a+1}$$

$$y_{zs}[n] = Z^{-1}\left\{\frac{Az}{z+a} + \frac{Bz}{z-\frac{1}{2}}\right\} = \frac{2a}{2a+1}(-a)^n u[n] + \frac{1}{2a+1}\left(\frac{1}{2}\right)^n u[n]$$

If $a = \frac{1}{2}$ then $Y_{zs}(z) = \dfrac{z^2}{(z-1/2)^2}$

$$\frac{Y_{zs}(z)}{z} = \frac{z}{(z-1/2)^2} = \frac{A}{(z-1/2)^2} + \frac{B}{(z-1/2)}$$

$$A = \left.z\right|_{z=1/2} = 1/2 \qquad B = \left.\frac{d}{dz}[z]\right|_{z=1/2} = 1$$

$$Y_{zs} = \frac{\frac{1}{2}z}{(z-\frac{1}{2})^2} + \frac{z}{z-\frac{1}{2}}$$

and

$$y_{zs}[n] = n\left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{2}\right)^n u[n]$$
$$= (n+1)\left(\frac{1}{2}\right)^n u[n]$$

- When A.C = 0 then $y_{zi}[n] = 0$ and $y[n] = y_{zs}[n]$

$$Y(z) = Y_{zs}(z) = H(z)\,X(z)$$

This allows us to easily determine the transfer function of a LCCDE description. and saves LCCDE $\rightarrow h[n] \rightarrow H(z)$ direct.

- Example: a second order system is given by

$$y[n] = \left(\tfrac{1}{4}\right) y[n-2] - \left(\tfrac{1}{8}\right) y[n-1] + x[n] - x[n-1]$$

with zero auxiliary conditions. What is the transfer function of the system?

- Taking Z transform

$$Y(z) = \left(\tfrac{1}{4}\right) z^{-2} Y(z) - \left(\tfrac{1}{8}\right) z^{-1} Y(z) + X(z) - z^{-1} X(z)$$

$$\left(1 + \tfrac{1}{8} z^{-1} - \tfrac{1}{4} z^{-2}\right) Y(z) = \left(1 - z^{-1}\right) X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 - z^{-1}}{1 + \frac{1}{8} z^{-1} - \frac{1}{4} z^{-2}} = \frac{z(z-1)}{z^2 + \frac{1}{8} z - \frac{1}{4}}$$

- Where do the A.C. come from? Suppose the previous example represents a filter. To compute $y[n]$ we need the previous two values of output $y[n-1]$ and $y[n-2]$ in addition to current and previous input $x[n]$, $x[n-1]$

- When we initialize the filter we have to choose the values in the output buffer and input buffer. These are the A.C.

- Thus it makes sense for us to just set them to zero. (zero initialize the arrays).

- Stability of LCCDE: To determine the stability we check poles of $H(z)$.

$$H(z) = \frac{\prod_{k=0}^{M} (z - \beta_k)}{\prod_{k=0}^{N} (z - \alpha_k)}$$

β_k are <u>zeros</u> of TF

α_k are <u>poles</u> of TF

the singularity of $H(z)$

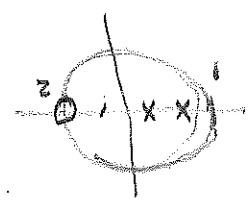
- A system is stable if $\forall k \; |\alpha_k| < 1$ (inside unit circle)

- A system is marginally stable if <u>non-repeated</u> poles on unit circle and remaining poles inside unit circle.

- A system is unstable if there are repeated poles on unit circle.

- Examples.

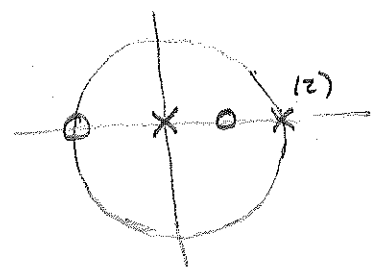
$$H(z) = \frac{K(z+1)^2}{(z-\frac{1}{2})(z-\frac{3}{4})}$$



stable.

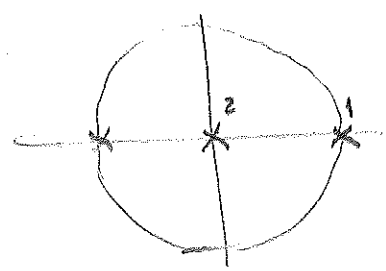
$$H(z) = \frac{K(z+1)(z-\frac{1}{2})}{z(z^2+2z+1)}$$

$$z(z-1)^2$$



unstable

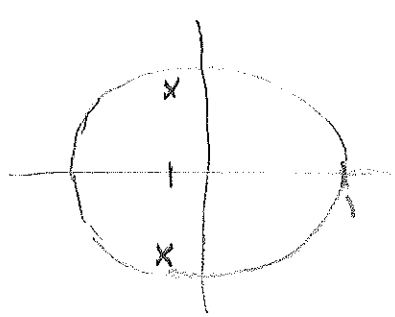
$$H(z) = \frac{z^2 + \frac{1}{2}z - 4}{z^2(z-1)(z+1)}$$



marginally stable

$$H(z) = \frac{1}{2z^2 + \frac{1}{2}z + 1}$$

$$\left(z + \frac{1}{8} + j\frac{\sqrt{31}}{8}\right)\left(z + \frac{1}{8} - j\frac{\sqrt{31}}{8}\right)$$



stable.

Jury Stability Criteria

- There is a similar shortcut to determining the stability of DT systems similar to the Routh-Horwitz criteria for CT systems.

- Let $H(z) = \dfrac{P(z)}{Q(z)}$ where $Q(z) = a_N z^N + a_{N-1} z^{N-1} + \cdots a_1 z + a_0$

Test 1: $Q(1) > 0$

Test 2: $(-1)^N Q(-1) > 0$

Test 3: $|a_0| < |a_N|$

If <u>any</u> of these test fail the system is unstable,

If all pass, then the final test uses the Jury array

row 1	a_0	a_1	$\cdots$		a_N
row 2	a_N	a_{N-1}	$\cdots$		a_0
row 3	b_0	b_1	$\cdots$		b_{N-1}
row 4	b_{N-1}	b_{N-2}	$\cdots$		b_0
row 5	c_0	c_1	$\cdots$		c_{N-2}
row 6	c_{N-2}	c_{N-3}	$\cdots$		c_0
	$\vdots$				
row 2N-3	v_0	v_1		v_3	

if the first and last column of <u>all</u> odd rows satisfys

$|b_0| > |b_{N-1}|$, $|c_0| > |c_{N-2}|$, etc, ... $|v_0| > |v_3|$

(Note: you can stop as soon as any test fails)

To find the values for rows > 2

$$b_k = \begin{vmatrix} a_0 & a_{N-k} \\ a_N & a_k \end{vmatrix} = a_0 a_k - a_N a_{N-k} \qquad k = 0, 1, \ldots N-1$$

$$c_k = \begin{vmatrix} b_0 & b_{N-1-k} \\ b_{N-1} & b_k \end{vmatrix} = b_0 b_k - b_{N-1} \cdot b_{N-1-k} \qquad k = 0, 1, \ldots N-2$$

continuing the pattern, ... Thus, the use of both rows.

- Example #1 $\quad H(z) = \dfrac{z^2 - z + 1}{4z^2 - 1}$

$$Q(z) = 4z^2 - 1 = \underset{a_2}{\underset{\shortparallel}{4}} z^2 + \underset{a_1}{\underset{\shortparallel}{0}} \cdot z - \underset{a_0}{\underset{\shortparallel}{1}}$$

$Q(1) = 4 - 1 = 3 > 0$ ✓

$(-1)^2 Q(-1) = 4 - 1 = 3 > 0$ ✓

$|-1| < |4|$ ✓

$\therefore$ stable

- Example #2 $\quad H(z) = \dfrac{1}{4z^2 + \frac{17}{2} z + 1}$

$$Q(z) = \underset{a_2}{\underset{\shortparallel}{4}} z^2 + \underset{a_1}{\underset{\shortparallel}{\tfrac{17}{2}}} z + \underset{a_0}{\underset{\shortparallel}{1}}$$

$Q(1) = 4 + \frac{17}{2} + 1 > 0$ ✓

$(-1)^2 Q(-1) = 4 - \frac{17}{2} + 1 = -\frac{7}{2} < 0$ ✗

$\therefore$ unstable

- Example #3 $\quad H(z) = \dfrac{3z^2 + z - \frac{1}{4}}{24z^3 + 26z^2 + 9z + 1}$

$Q(1) = 60 > 0$ ✓

$(-1)^3 Q(-1) = 6 > 0$ ✓

$|1| < |24|$ ✓

$b_0 = 1^2 - 24 \cdot 26 = \boxed{-623}$

$b_1 = 1 \cdot 9 - 24 \cdot 9 = -207$

$b_2 = 1 \cdot 26 - 24 \cdot 9 = \boxed{-190}$

1	9	26	24
24	26	9	1
b_0	b_1	b_2	

$\therefore$ stable

- As in CT we have to take care when pole/zero cancellation occurs, but now it is due to quantization effects in filter coefficients.

- Relationship of Z-transform to DTFT

Recall for a causal system $H(z)$ $|z| > r$

If the system is in addition stable then all poles inside unit circle and $r < 1$. Thus the ROC includes the unit circle $z = e^{j\omega}$ $\omega \in \mathbb{R}$. and

$$H(e^{j\omega}) = H(z)\Big|_{z=e^{j\omega}} \quad \text{for stable, causal systems.}$$

- As in CT, DT analysis typically proceeds as

System description $\longrightarrow H(z) \xrightarrow{\text{stable?}} H(\omega) \longrightarrow$ Fourier Analysis.

$h[n]$, LCCDE

B, D

$H(z) \xrightarrow{\text{unstable}}$ Z-transform (easiest when both input & system causal)

- Inverse Systems. As with Laplace the z-transform gives us a principled approach to finding the inverse of a system.

$$x[n] \longrightarrow \boxed{H_1(z)} \longrightarrow \boxed{H_2(z)} \longrightarrow y[n] = x[n]$$

implies $H_2(z) = \dfrac{1}{H_1(z)}$ if $H_1(z) = \dfrac{P(z)}{Q(z)}$

$= H_1^{-1}(z)$ $\qquad H_2(z) = \dfrac{Q(z)}{P(z)}$

NOTE zeros of H_1 become poles of H_2

thus the inverse is stable only if zeros of H_1 inside unit circle.